

# Comparing Strong and Weak Acids at the Particle Level

PhET Acid-Base Solutions · PhET

Senior Secondary (Years 11-12)

~30 minutes

Science · Chemistry

## NOTICE — AI-generated worksheet

This worksheet was drafted automatically from the site's curriculum-mapping data and has not been checked by a human curriculum expert. Please review every task for accuracy, safety and suitability for your specific class before use.

Simulation: <https://phet.colorado.edu/en/simulations/acid-base-solutions>

Curriculum links: acids and bases · equilibrium · solutions

## Learning intentions

- I can compare the particle-level behaviour of strong and weak acids at equal concentrations.
- I can relate the proportion of dissociated particles to the measured pH.
- I can explain acid strength in terms of equilibrium, rather than concentration.

## Getting started

Open PhET Acid-Base Solutions and select the comparison or dual-view mode, if available, so you can view a strong acid and a weak acid side-by-side at the same concentration. Note the pH meter and particle view for each.

## Tasks

1. Set up a strong acid and a weak acid at the same concentration. Record the pH shown for each.

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2. Look closely at the particle view for each solution. Describe the difference in what you see between the strong acid and the weak acid.

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3. Increase the concentration of the weak acid. Does its pH become equal to that of the strong acid at the original concentration? Explain what you observe.

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4. Compare the strong acid and weak acid at two different concentrations each, recording pH and the approximate proportion of dissociated particles.

| Acid type | Concentration | pH | Approx. proportion dissociated |
|-----------|---------------|----|--------------------------------|
|           |               |    |                                |
|           |               |    |                                |
|           |               |    |                                |
|           |               |    |                                |

5. Explain why the weak acid's particle view shows a mixture of dissociated ions and intact molecules, while the strong acid's does not, or only in trace amounts.

6. If a strong acid solution has a hydrogen ion concentration of  $1 \times 10^{-2}$  mol/L, calculate its pH. Compare this to the pH shown in the simulation for a strong acid at a similar concentration.

Working:

7. Explain, using the term 'equilibrium', why acid strength is not the same thing as acid concentration.

**Extension (optional)**  
Predict what would happen to the pH and particle view if you diluted the weak acid solution by a factor of 10, then test your prediction. Explain any result that surprised you, referring to Le Chatelier's principle if relevant.